Equipment Qualification

Work Instruction

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Mastercard CQMAP V3.A — NEXUS Qualification Hub

Work Instruction — Qualifying New Production Equipment in NEXUS

Document ID: WI-NEXUS-EQUIP-001 **System:** Quality Control Hub — NEXUS Qualification Hub **Standard:** Mastercard CQMAP V3.A (CQM Requirements 3a, Nov 2025) **Audience:** Quality technicians and engineers with no prior NEXUS experience

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1. Purpose

This work instruction walks you through what to do in NEXUS when a **new machine** arrives on the production floor — for example, a new punch, laminator, embosser, or bonder — and needs to be **qualified** before it can be used to make product that ships.
You do not need to already know what "Cpk," "#0706#," or "conformity" mean. Section 4 explains every term the first time it comes up. If you only remember one thing from this document, remember this:

> **A new machine cannot make shippable product until its qualification plan's gate shows PASSED.** Until then, keep production on the old, already-qualified machine.

2. Scope

Use this work instruction any time a **physical piece of production or test equipment** is installed, replaced, or significantly modified — not for qualifying a new card design (that has its own work instruction, `WI-`

`NEXUS-QUAL-001`). This covers Site Acceptance Testing (SAT) and the equipment qualification checklist that follows it.

3. Roles & responsibilities

Role	Responsibility
Quality Manager	Owns the qualification plan; assigns who does each checklist item; decides when the plan is promoted to `approved`.
Process Engineer / Facilities	Confirms the machine is installed correctly (IQ) and provides the Factory Acceptance Test (FAT) report.
Quality Engineer / Technician (you)	Runs the first-article cards through dimensional checks, enters results, keeps the checklist up to date.
Quality Manager (Reviewer)	Signs off the intermediate and final design reviews before production can switch to the new machine.

4. Key terms

Term	Plain-language meaning
SAT (Site Acceptance Test)	Proving the machine works correctly *here, in your building*, after installation — as opposed to a **FAT** (Factory Acceptance Test), which the equipment vendor runs at *their* factory before shipping it to you.
IQ (Installation Qualification)	A documented check that the machine is physically installed correctly — power, air, placement — before you try to run product on it.
First article	The first batch of real cards run on the new machine, set aside specifically to be measured and checked before you trust the machine with normal production.
Cpk	A single number that says how consistently a machine hits its target dimension. `Cpk ≥ 1.33` is the bar NEXUS requires — below that, the machine is not considered "in control" yet.
Conformity (`Full` / `RI` / `nc-` / `NC+`)	The finding for one line item. `Full` = meets requirement. `RI` = "Requires Improvement" — not yet meeting the bar but not a hard failure. `nc-` / `NC+` = a real non-conformity, worse as it goes toward `NC+`. A brand-new machine under qualification normally sits at `RI` until it passes.
`#0706#` gate	NEXUS's shorthand for the rule "only ship product made on a qualified process." Every qualification plan has a gate panel that is either **NOT PASSED** (production stays on the old machine) or **PASSED** (the new machine may be used for shippable product).
Product scope	Which CQM product category (e.g. `CB` — Card Body, `ICC` — IC Card) the plan is qualifying. If your new machine touches more than one product category, you need **one plan per category** — the gate is evaluated per product, not once for the whole site.

5. The process at a glance

```
Machine physically installed
        │
        ▼
IQ (install/utilities) + FAT report obtained  ──►  document in the checklist
        │
        ▼
Operator training + work instructions updated
        │
        ▼
First-article run on the new machine
        │
        ▼
Dimensional checks (ISO 7810: width/height, corners, edges, warpage…)
        │
        ▼
Intermediate design review          [checkpoint]
        │
        ▼
Cpk capability study (≥ 1.33, ~30 cards)
        │
        ▼
Final design review — must be APPROVED
        │
        ▼
#0706# GATE — all conditions green ──► new machine may be used for production
```

Every box above is one or more rows on the qualification-plan checklist you will build in Section 7. Nothing here is optional or reorderable — the gate will not pass until every step is done.

6. Prerequisites

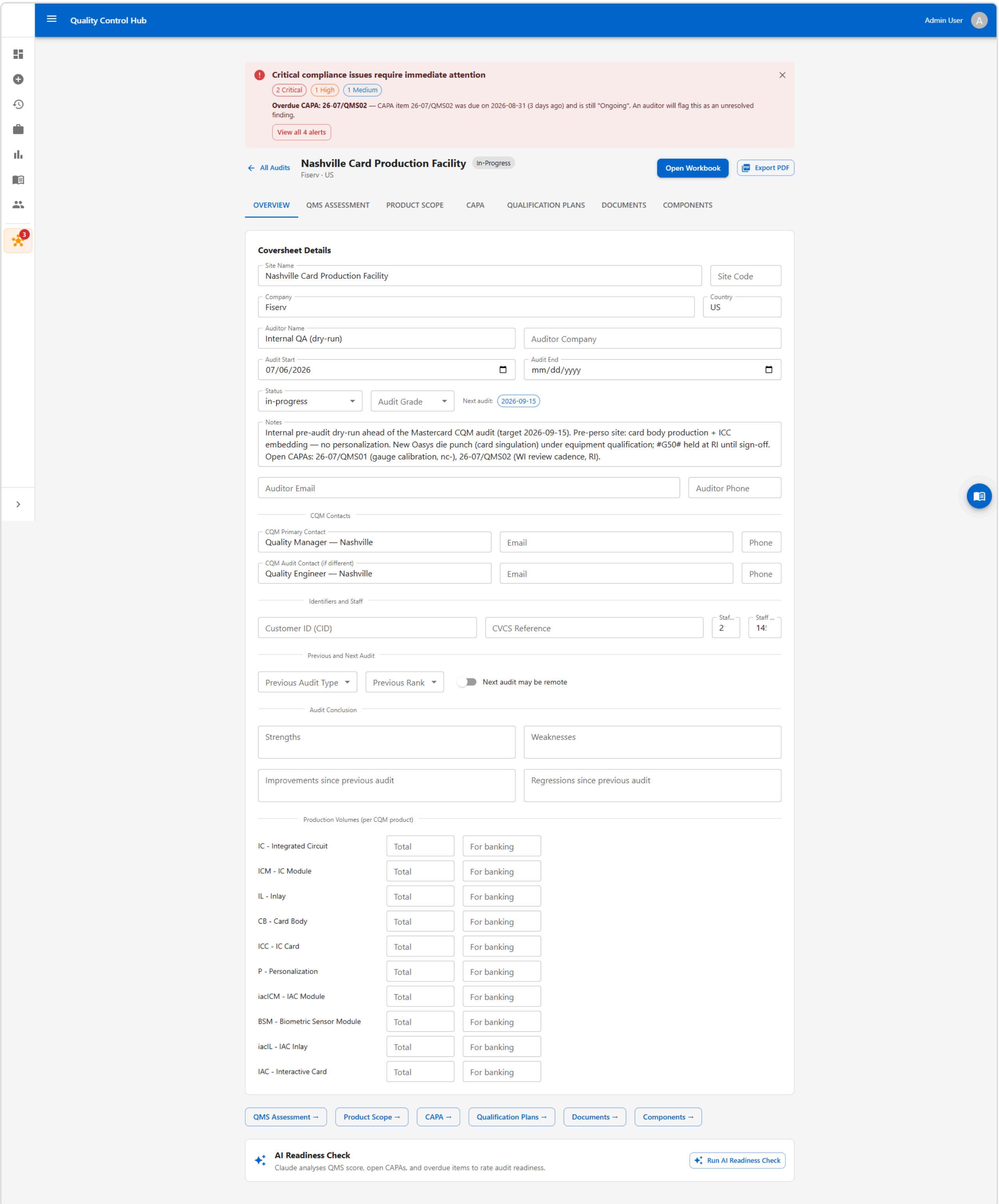
Before you start:

- You are signed in to the Quality Control Hub with a **Quality Manager**, **Auditor**, or **Admin** role.
- A NEXUS **audit record** already exists for your site (ask your Quality Manager if you are not sure — do not create a new one just for an equipment qualification).
- You know which CQM **product scope(s)** the new machine affects (e.g. it cuts card bodies used in both `CB` and `ICC` products). It is fine if those scopes have not been added to NEXUS yet — Step 2 below covers that.

7. Step-by-step procedure

Step 1 — Open your site's audit record

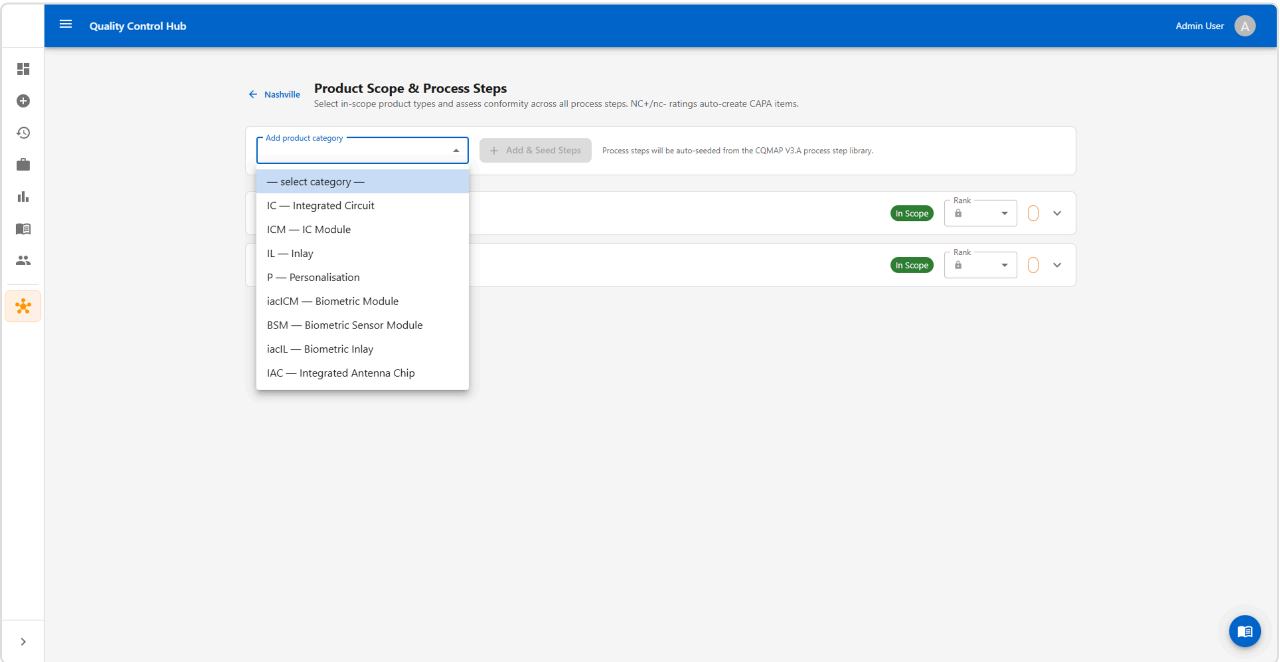
From the left sidebar, open **NEXUS Hub**, then click into your facility's audit record.



> Notice the alert banner at the top — NEXUS automatically watches for overdue items and upcoming audits and puts the most urgent one right at the top. Check this banner every time you open the audit; it will tell you if something related to your equipment qualification has gone overdue.

Step 2 — Make sure your product scope exists

A qualification plan has to be attached to a **product scope** (a CQM product category like `CB` — Card Body or `ICC` — IC Card) — the `#0706#` gate checks conformity against that specific product's process steps, so a plan cannot be usefully created without one. If your site's product scopes are already set up, skip to Step 3. Open the audit record's **PRODUCT SCOPE** tab. Use the **Add product category** dropdown to pick the category your machine affects, then click **Add & Seed Steps**.

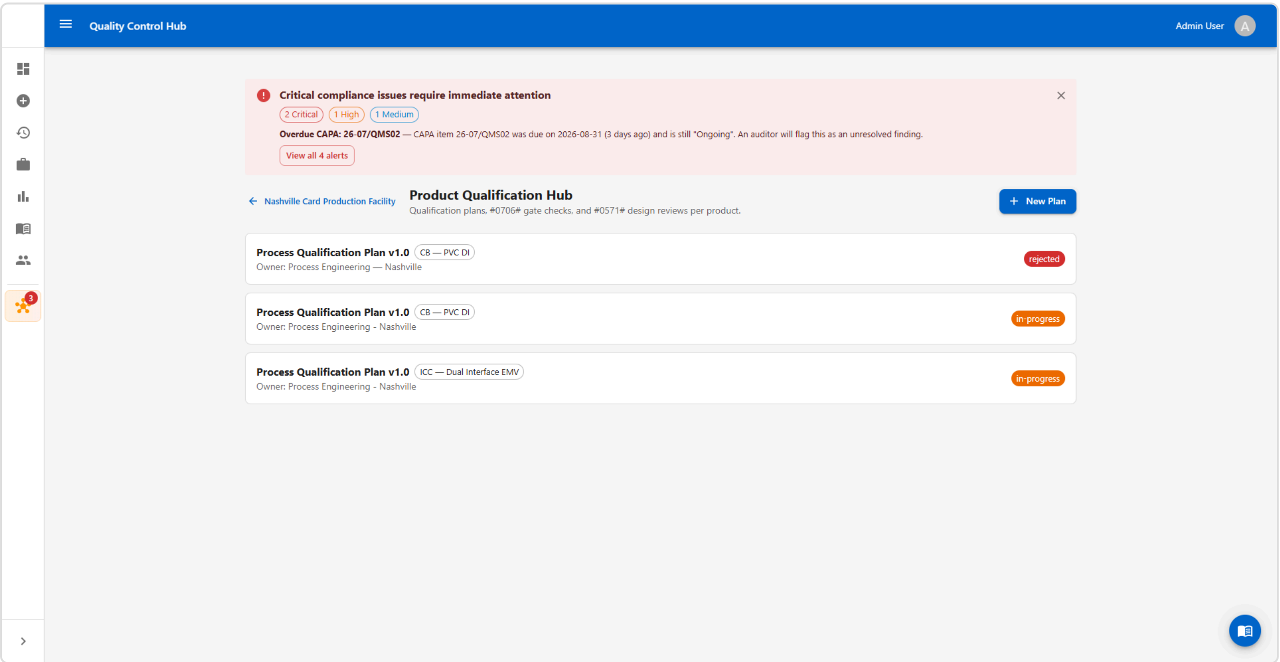


Repeat once for every product category the machine feeds (a card-cutting punch, for example, typically touches both `CB` and `ICC`). Each category you add auto-seeds roughly 90–120 process steps from the CQMAP V3.A library — you do not fill these in now, only the one your machine performs, later in Step 6.

> If this step is skipped, the **Product scope** dropdown in Step 4's "New Plan" dialog will be empty except for "None (audit-level)" — and a plan created with no scope can never pass its gate. If you already made that mistake, the fix is a two-line database update, not starting over — ask whoever set up NEXUS for you to link the existing plan to the correct scope once it exists.

Step 3 — Open Qualification Plans

Click the **QUALIFICATION PLANS** tab. You will see every qualification plan for this site, newest first, each tagged with the product scope it covers and its current status (`in-progress`, `approved`, `rejected`, etc.).



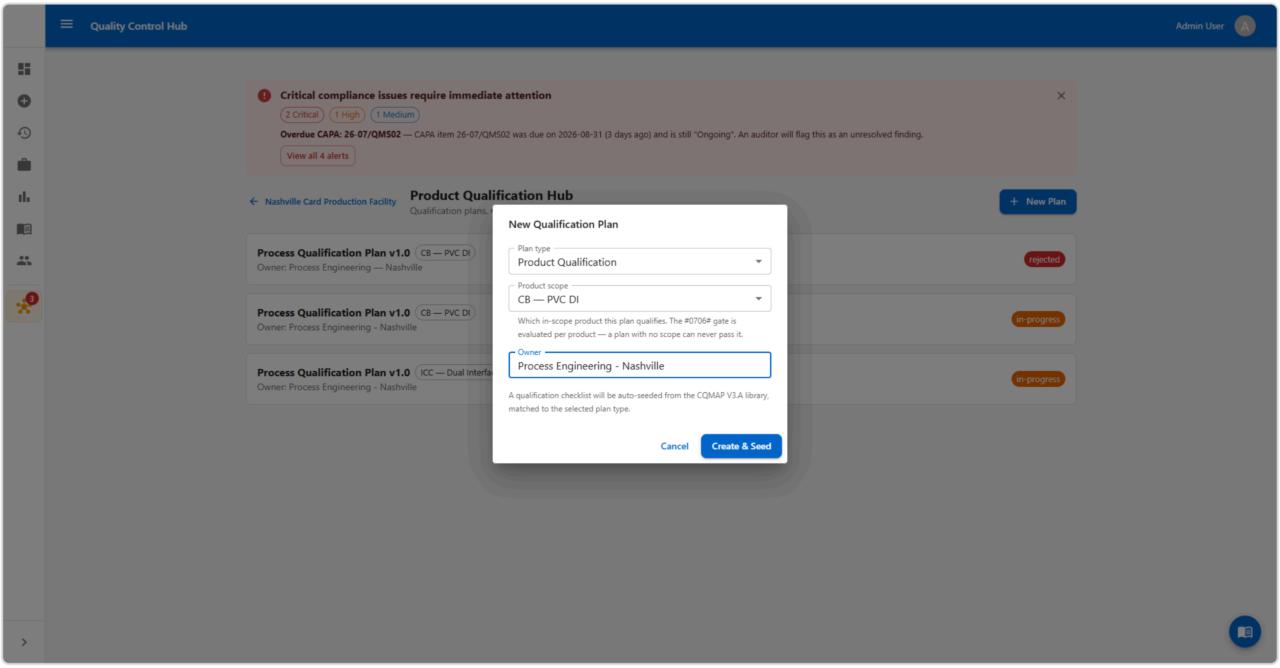
> If an old plan exists for the same machine with dates that clearly did not happen (for example, it says the machine was installed on a date you know is wrong), do not edit it to "fix" the dates. Leave it as a record, set its status to `rejected`, and create a fresh plan instead — this keeps an honest history of what actually happened. Ask your Quality Manager if you are unsure whether an existing plan applies to your machine.

Step 4 — Create one plan per affected product scope

Click **New Plan**. Fill in:

1. **Plan type** — choose **Process Qualification** (you are proving the *process on this machine* is capable, not redesigning the *card*).
2. **Product scope** — pick the product this plan covers, from the categories you added in Step 2. **If the machine feeds more than one product category, repeat this whole step-by-step procedure once per category** — do not try to cover two products with one plan.
3. **Owner** — the Quality Manager responsible for driving this plan to completion.

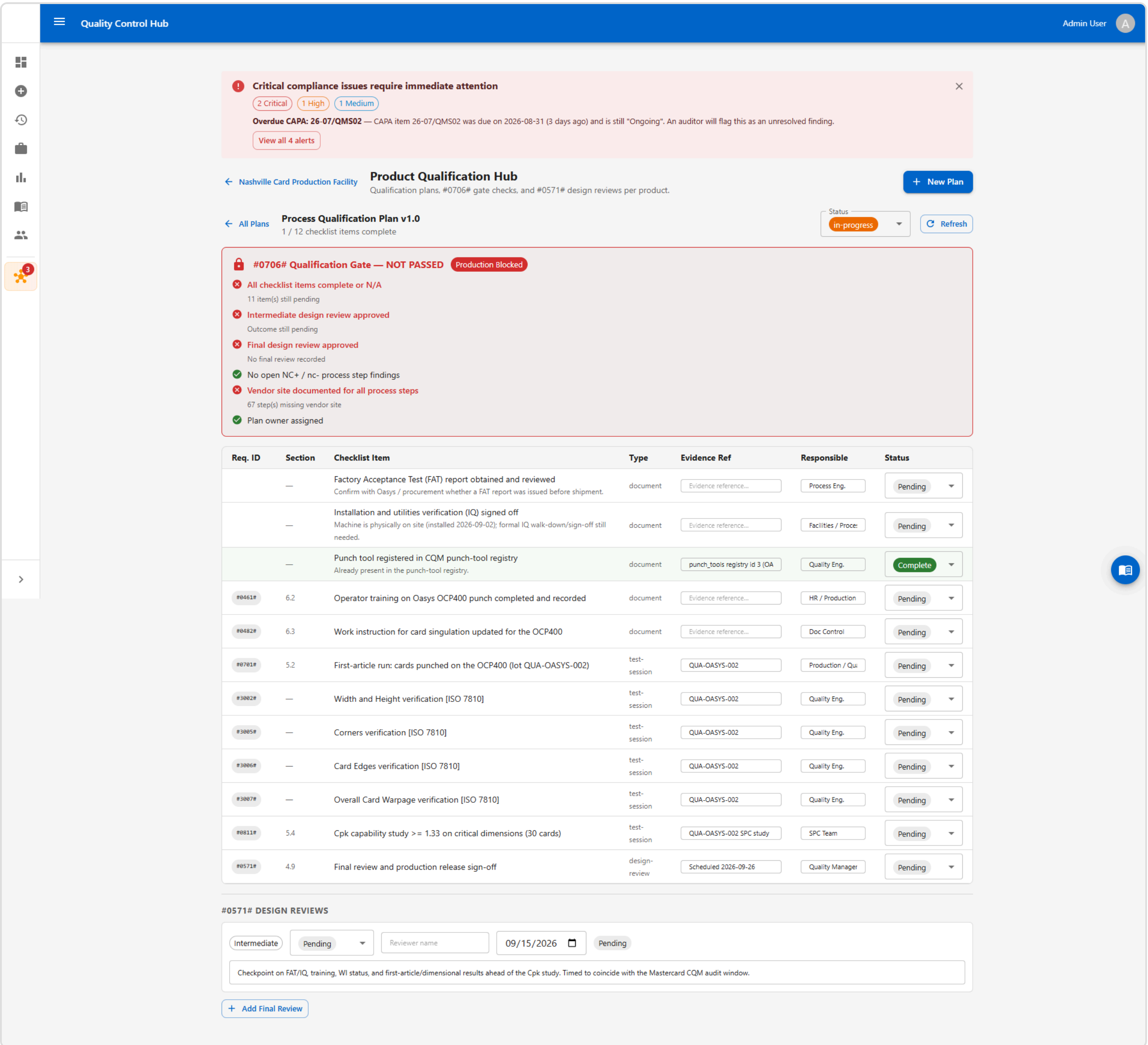
Click **Create & Seed**.



> The helper text under "Product scope" is not a suggestion — a plan with no scope selected can **never** pass its gate, because the gate has nothing to check conformity against. Always pick a scope. If the dropdown is empty, you skipped Step 2 — go do it now.

Step 5 — Work the checklist

Open the plan you just created. You will see the gate panel at the top (still red — that is expected right now) and the checklist below it.



For each row:

- **Evidence Ref** — type in a document number, lot code, or reference proving the work happened (e.g. a FAT report number, or the lot code for your first-article cards).
- **Responsible** — who owns getting this specific line done.
- **Status** — leave `Pending` until real evidence exists, then move it to `In Progress` and finally `Complete`. **Never mark something complete before it has actually happened** — the gate exists to stop unqualified product, and a false "complete" defeats that purpose.

A typical equipment-qualification checklist looks like this, in order:

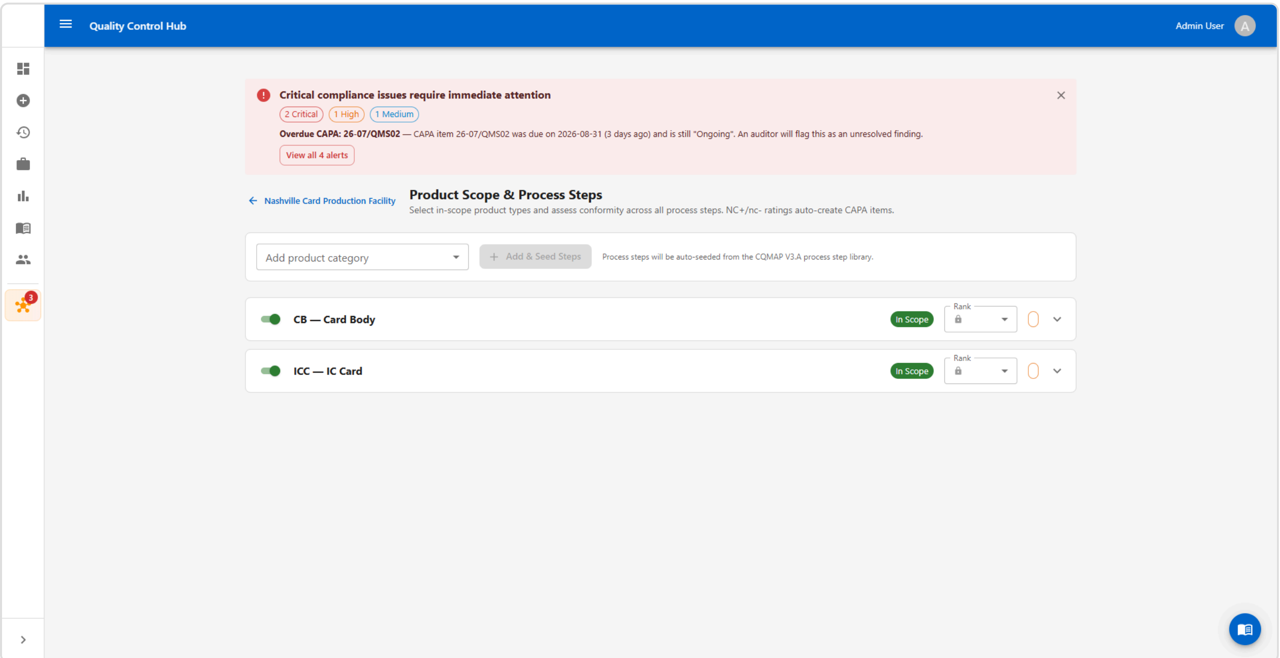
1. Factory Acceptance Test (FAT) report obtained and reviewed
2. Installation and utilities verification (IQ) signed off
3. Punch/production tool registered in the CQM tool registry
4. Operator training completed and recorded
5. Work instruction updated for the new machine
6. First-article run completed (note the lot code)

7. Dimensional checks per ISO 7810 (width/height, corners, edges, warpage — whichever apply to what this machine makes)
8. Cpk capability study, ≥ 1.33, on at least ~30 cards
9. Final review and production release sign-off

Step 6 — Note the machine on the affected process step

Go back to the **PRODUCT SCOPE** tab you used in Step 2, then expand the product category your plan covers. Find the process step your new machine performs (for a card-cutting/punch machine, this is `#G50# Card Singulation`) and fill in:

- **Vendor Site** — your facility name/location.
- The conformity dropdown should read `RI` while qualification is in progress — this is correct and expected, not a mistake to fix.



There is a free-text notes area on this row (open the row detail) — use it to record which machine is doing this step now, e.g. "*New \[machine name/model] installed \[date] — qualification in progress; production remains on \[old machine] until sign-off.*" This is what an auditor reads to understand what is happening on your floor right now.

Step 7 — Record the design reviews

Still on the plan detail page, scroll to **Design Reviews**. Add an **Intermediate Review** once training, work instructions, and first-article results are in — this is a checkpoint, not the finish line. Later, once the Cpk study is done and everything looks good, add the **Final Review** and mark its outcome `approved`.

> The gate needs the intermediate review to be `approved` **or** `conditional`, and the final review to be

> `approved` — no other outcome will open the gate.

Step 8 — Watch the gate, do not rush it

The gate panel checks six things in real time:

1. All checklist items complete or N/A
2. Intermediate design review approved (or conditional)
3. Final design review approved
4. No open `NC+` / `nc-` findings on the process step
5. Vendor site filled in for the process step
6. Plan owner assigned

It is normal and expected for the gate to stay NOT PASSED for several weeks. A real equipment qualification — training, a first-article run, dimensional checks, and a proper Cpk study — takes time. Do not shortcut steps to make the gate pass faster; that is exactly the failure mode this checklist exists to prevent. If a Mastercard audit happens to land while the gate is still open, that is fine — present the plan as an active, documented qualification and keep production on the already-qualified machine in the meantime.

Step 9 — Promote the plan status as work finishes

Update the plan's **Status** dropdown as things progress: `draft → in-progress → submitted → approved` (or `rejected` if the machine fails qualification). Only set it to **approved** once the gate panel shows all six conditions green.

8. Completion criteria

The equipment qualification is done when:

- The gate panel shows **all six conditions passed** — no red "Production Blocked" banner.
- The final design review is recorded with outcome **approved**.
- The plan status is set to **approved**.
- The process step's conformity has been updated from `RI` to `Full`.

Only then should production switch from the old machine to the new one for shippable product.

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9. Worked example — Oasys OCP400 die punch

This example is a real plan built in NEXUS, referenced throughout the screenshots above, so you can see exactly what a completed setup looks like in practice.
Situation: A new Oasys OCP400 high-speed punch (card singulation — cuts finished cards out of laminated sheets, ±0.1mm tolerance, up to 60,000 cards/hour) was installed on the floor to eventually replace an older

punch. It affects two product categories: `CB` (card body) and `ICC` (finished IC cards).

What was set up:

- The `CB` and `ICC` product scopes (Step 2) — on a freshly created audit record, neither existed yet, so the "Product scope" dropdown in the New Plan dialog was empty until these were added.
- Two Process Qualification plans — one for `CB`, one for `ICC` — because the `#0706#` gate is evaluated per product, and a machine feeding two product lines needs two plans.
- A 9-line checklist on each plan: FAT report, IQ sign-off, tool registration, operator training, work instruction update, first-article run, four ISO 7810 dimensional checks, a Cpk study, and final sign-off.
- The `#G50#` process step on both product scopes updated to show the new machine, `RI` conformity, and the site name.
- An intermediate design review scheduled as a checkpoint.

What was deliberately left alone: only the checklist line that was independently verifiable (the tool registry entry) was marked complete. Everything else — training, the FAT report, first-article results, the Cpk study — was left `Pending` with a realistic target date, because none of it had actually happened yet. **Do the same on your own machine: only mark a line complete when you can point to real evidence.**

10. Reference

- `docs/qualification-work-instructions.md` (WI-NEXUS-QUAL-001) — the related work instruction for qualifying a *card product/design*, as opposed to a *machine*.
- `docs/nexus-qualification-hub.md` — how the NEXUS Qualification Hub maps to the Mastercard CQMAP V3.A requirements catalog.

11. Revision history

Version	Date	Author	Notes
1.0	2026-09-03	Quality Control Hub	Initial issue, written against the Oasys OCP400 die-punch equipment qualification.
1.1	2026-09-03	Quality Control Hub	Added Step 2 (create the product scope before creating a plan) — the original issue assumed product scopes already existed and did not explain how to add them, which left the "New Plan" dialog's Product scope dropdown empty on a freshly created audit.